

BTEC Level 3 Applied Science · Unit 1 Biology · Answer sheet

Prokaryotic and Eukaryotic Cells — Answers

Mark scheme for the companion worksheet. Accept equivalent correct wording. These are original JDSscience items, not official Pearson questions.

A–E

1. Eukaryotes have a nucleus / membrane-bound organelles; prokaryotes do not.
2. Cell wall, membrane, loop of DNA, ribosomes; also plasmid / flagellum / capsule.
3. Neither — they are acellular / not cells.
4. Accept correct labels from the schematic.
5. Nucleus absent/present; circular vs linear DNA; no/yes membrane-bound organelles; 70S vs 80S; 1–5 µm vs 10–100 µm; peptidoglycan vs cellulose/chitin/none.
6. Small extra DNA circle; may carry resistance genes or be used as a vector.
7. Antibiotics can target peptidoglycan walls or 70S ribosomes, which human cells lack or have in a different form.
8. Three paired differences. [3]
9. Carry resistance genes; transferable; used as vectors. [3]
10. No nucleus / DNA not enclosed by a nuclear envelope / no membrane-bound organelles. [2]
11. These features resemble free-living bacteria, supporting an engulfed-prokaryote origin.